IV. Give detailed answers to the following questions

40. Explain the method of least squares

41. The following data states the number of ATM centers during 1995 to2001.

Year	1995	1996	1997	1998	1999	2000	2001
Number of ATM centres	50	63	75	100	109	120	135

Obtain the trend values using semi averages method

42. From the following data estimate the trend values using semi averages method

Year	2003	2004	2005	2006	2007	2008	2009	2010
Consumption of cotton (Thousands of bales)	677	696	747	755	766	777	785	836

43. Following data gives the yield of food grains in India for the years 2000-01 to 2009-10. Find the trend values using 4 year moving averages.

Year	2000-01	2001-02	2002-03	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10
Yield (kg/ hectare)	1626	1734	1535	1727	1652	1715	1756	1860	1909	1798

44. Estimate the value of production for the year 1995 by using the method of least squares from the following data.

Year	1990	1991	1992	1993	1994
Production (1000s tons)	70	72	88	90	92

45. Find the following for the calculation of number of telephones for the year 2000.
    (1) Fit a straight line trend by the method of least squares.
    (2) Calculate the trend values.

Year	1990	1991	1992	1993	1994	1995
No. of telephones (in ^{1}00s)	20	21	23	25	27	29

46. The following data describes the export quantity of a company.

Year	1995	1996	1997	1998	1999	2000	2001
Export (in millions)	12	13	13	16	16	19	23

Fit a straight line trend and estimate the export for the year 2005.

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